

EXPERIMENT NO. 11:

Write a program for the task of Credit Score Classification

Aim:

To implement a Credit Score Classification program in Python to classify customers into different credit score categories based on their financial attributes.

Procedure:

1. Import the required Python libraries.
2. Create a sample credit score dataset.
3. Separate the input features and target classes.
4. Split the dataset into training and testing sets.
5. Train a classification model using the training data.
6. Predict the credit score category for the test data.
7. Calculate the accuracy of the model.
8. Classify a new customer's credit score category.

Program:

Credit Score Classification

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

# Sample Credit Score Dataset
data = {
    'Income': [25000, 30000, 45000, 50000, 60000, 70000,
               80000, 90000, 35000, 40000, 55000, 75000],
    'Age': [22, 25, 28, 30, 35, 40, 45, 50, 26, 32, 38, 42],
    'Credit_History': [1, 1, 0, 1, 1, 1, 1, 1, 0, 1, 1, 1],
    'Loan_Defaults': [3, 2, 4, 1, 0, 0, 0, 0, 3, 1, 0, 0],
    'Credit_Score': [
        'Poor', 'Poor', 'Poor', 'Good', 'Good', 'Good',
```

```

        'Good', 'Good', 'Poor', 'Good', 'Good', 'Good'
    ]
}

df = pd.DataFrame(data)

# Input features and target
X = df[['Income', 'Age', 'Credit_History', 'Loan_Defaults']]
y = df['Credit_Score']

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)

# Create Decision Tree Classifier
model = DecisionTreeClassifier(random_state=42)

# Train the model
model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Actual Values:")
print(y_test.values)

print("\nPredicted Values:")

```

```
print(y_pred)
```

```
print("\nAccuracy:")
```

```
print(round(accuracy * 100, 2), "%")
```

```
# Predict a new customer's credit score
```

```
new_customer = [[65000, 36, 1, 0]]
```

```
prediction = model.predict(new_customer)
```

```
print("\nNew Customer Credit Score:")
```

```
print(prediction[0])
```

```
""" Actual Values:
    ['Good' 'Good' 'Poor']

    Predicted Values:
    ['Good' 'Poor' 'Poor']

    Accuracy:
    66.67 %

    New Customer Credit Score:
    Good
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but DecisionTreeClassifier was fitted with feature names
  warnings.warn(
```

Result:

The Credit Score Classification program was successfully implemented in Python using a Decision Tree Classifier. The model classified customers based on income, age, credit history, and loan defaults, and successfully predicted the credit score category of a new customer.